D.Y. PATIL COLLEGE OF ENGINEERING & TECHNOLOGY,

KASABA BAWADA, KOLHAPUR

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

(2024-25)

Project-II

Report on

"Hall Ticket Generator with Barcode"

Submitted By

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2	Swayam Dakare	67
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Under the guidance of

Prof. T. V. Salokhe

Class: TY (CSE) **Div: A** **Batch: T3**

D.Y. PATIL COLLEGE OF ENGINEERING AND TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

KOLHAPUR

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COLLEGE of
ENGINEERING & TECHNOLOGY
(AN AUTONOMOUS INSTITUTE)
KASABA BAWADA, KOLHAPUR

CERTIFICATE

This is to certify that the project-II group consisting the following members has satisfactorily completed the Project-II work entitled "Hall Ticket Generator with Barcode" at TY (CSE) semester V as prescribed in the syllabus for the academic year 2024-2025

Name	Exam seat no.
1. Soham Bagade	
2. Swayam Dakare	
3. Shridhar Badger	
4. Ketan Patil	

Date:

Place: Kolhapur

(Project Guide) **(HOD)**

(Project coordinator) **(Principal)**

External Examiner

ACKNOWLEDGEMENT

This project work entitled “Hall Ticket Generator with Barcode” was a formidable task, but with collective effort of our group and active guidance made it possible for us to complete.

It is our pleasure to acknowledge the help we have been received from institute and the Individual. We would like to thank our Principal **Dr. Santosh Chede** Principal, D. Y. Patil College of Engineering and Technology, Kolhapur in particular for always giving encouragement, support and the excellent facilities provided.

We would like to thank Prof. Miss. Raddhika. J. Dhannal (H.O.D., Department ofcomputer science and engineering) delineating us with this project work.

We would also like to thank **him/her** for the support and interest that **he/She** has shown in bringing out this project and our course co-coordinator Prof. Nandini Patil for **his/her** guidance and cooperation We would also like to express our most humble and deepest gratitude to Project Guide Tushar. V. Salokhe, for providing us with the right guidance at the time of need itwas for **his/her** presence and active guidance that we were able to complete the project work.

We would like to thank all our friends for their help, ideas, criticisms and also their encouragements for preparation of this project work. Any further ideas and constructive criticisms on our work shall be highly welcomed.

Name **Sign**

1. Soham Bagade
2. Swayam Dakare
3. Shridhar Badger
4. Ketan Patil

Date:

Place: Kolhapur

Abstract

The project aims to develop a comprehensive Hall Ticket Generator system using Python, integrated with Barcodes for secure and efficient verification. This system is designed to address the challenges faced by educational institutions in manually creating and distributing hall tickets. By automating this process, the system enhances accuracy, reduces the risk of forgery, and simplifies the verification process at examination centres. The inclusion of Barcodes further ensures that each hall ticket is unique and can be quickly validated, thus streamlining the entire examination management process.

This project implements a barcode-based hall ticket generation system, following the approach of Sarjiyus et al. (2024), to secure and streamline the process of student verification during examinations [1].

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1. Introduction

Hall tickets are a critical component of the examination process in educational institutions, servingas a student's entry pass to the exam hall. The traditional approach to generating these tickets involves manual data entry and printing, which can lead to significant delays, human errors, and potential security breaches such as ticket forgery. Moreover, the manual verification of hall ticketsat exam centers is time-consuming and inefficient. This project proposes an automated system thatleverages Python's capabilities to generate hall tickets, embedding Barcodes for added security.

2. Need of Work

The development of the Hall Ticket Generator with Barcode addresses the growing need for efficient, accurate, and scalable solutions in academic and professional examination management. As educational institutions and examination bodies strive to improve their administrative processes, there is a pressing need for systems that automate hall ticket generation while ensuring security and accuracy. This project is aimed at:

- **Error Elimination and Efficiency:**
  Automating the process of hall ticket generation to reduce human errors, improve accuracy, and save time. By eliminating manual data entry, this project ensures error-free hall ticket creation for large-scale student data.

- **Secure Identification and Verification:**
  Integrating barcodes for quick and reliable student identification during examinations. Barcodes improve security by preventing unauthorized access and streamlining the verification process.

- **Scalability and Accessibility:**
  Providing a solution that can handle data for hundreds or thousands of students with ease. This ensures that institutions can generate hall tickets and manage student information regardless of the batch size.

3. Problem Statement

To improve the management of student data and hall ticket generation, educational institutions need an automated system. The current manual processes lead to errors, duplicate entries, and slow verification at examination centers, resulting in long queues and increased risks of unauthorized access. A secure, efficient solution is essential to uphold the integrity of the examination process.

4. Objectives:

The primary objectives of this project are:

- Automation: To develop a system that automates the generation of hall tickets, minimizing manual intervention and errors.
- Security: To integrate barcodes into hall tickets, ensuring each ticket is unique and can be securely verified.
- Efficiency: To reduce the time and resources required for generating and verifying hall tickets, streamlining the overall examination management process.
- Scalability: To design a system that can handle large volumes of data and be easily scaled to accommodate different institutions and examination formats.
- User-friendliness: To provide an intuitive interface for administrators to manage student data and generate tickets without requiring advanced technical skills.
- Reduce Manual Efforts: The system minimizes human error and paperwork involved in generating and distributing hall tickets [2].

5. Methodology (Diagram with Explanation):

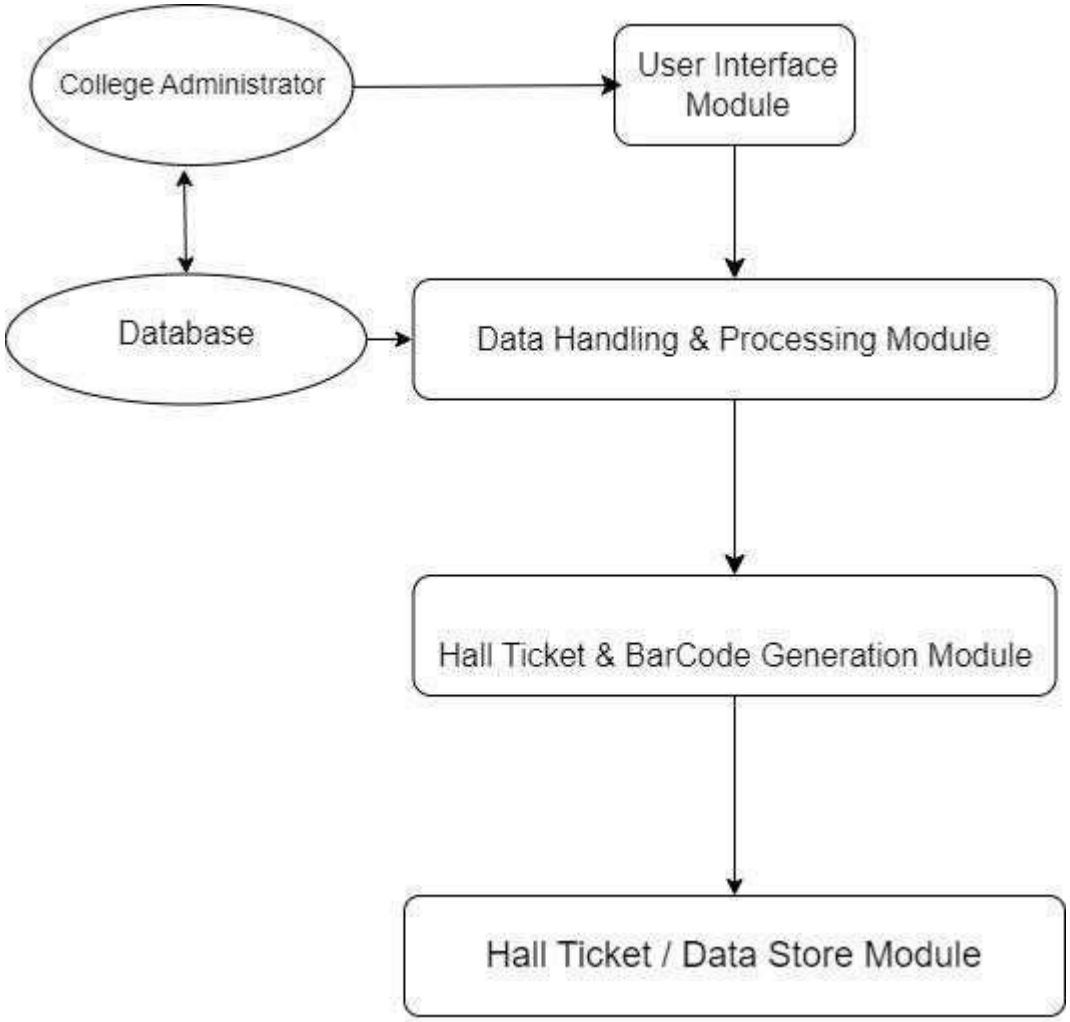


Figure 5.1 Data Flow Diagram of Hall Ticket Generator with Barcode

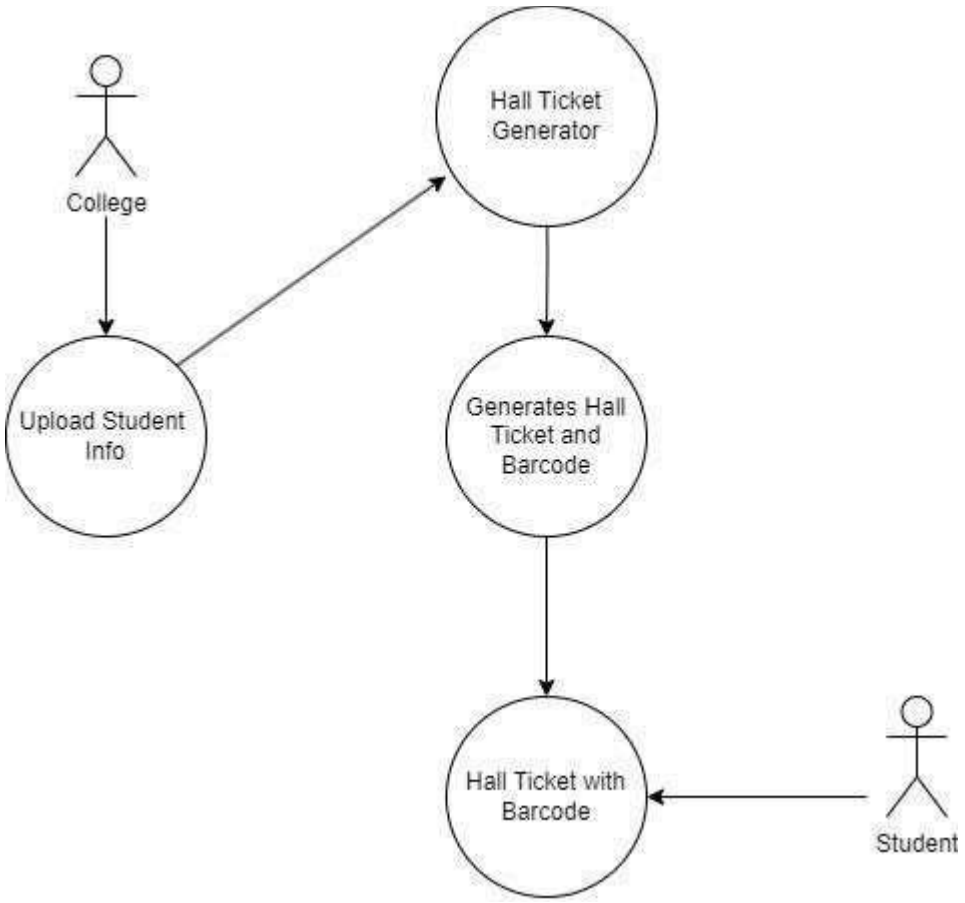


Figure 5.2 Use case Diagram for Hall Ticket Generator with Barcode

6. Algorithm/Pseudocode:

Purpose: The purpose of the algorithm is to automate the generation of hall tickets with barcodes and student details using data from an Excel file through a user-friendly GUI.

Step 1: Start

Step 2: Define generate barcode(student, output path) to:

Create and save a barcode image for the student's unique ID.

Step 3: Define create_hall_ticket(student, output path) to:

Create a landscape PDF with:

College name and title at the top.

Student details and timetable in the body.

Barcode in the top-right corner.

Space for Principal's signature at the bottom-right corner.

Save the PDF and delete the temporary barcode image.

Step 4: Define process_file(file_path) to:

Load student data from an Excel file.

Iterate through each record and call create_hall_ticket.

Print success messages for each generated hall ticket.

Step 5: Define upload file() to:

Open a file dialog for selecting an Excel file.

Pass the file path to process_file.

Step 6: Define main() to:

Create a GUI with a label and button for file upload.

Start the GUI event loop.

Step 7: Call main() to run the program.

Step 8: End

Output: A program to generate hall tickets with student details, timetable, barcode, and principal's signature from an Excel file via a GUI for file upload.

7. Modules:

The system will be composed of several interconnected modules, each responsible for a specificfunction:

- **User Interface Module:** This module will provide a web-based or desktop application interface for administrators. The UI will be designed to be intuitive and easy to navigate,allowing for easy input, management, and tracking of student data.
- **Data Handling and Processing Module:** Responsible for processing the input data, this module will validate the data to ensure accuracy and completeness before generating thehall tickets. It will also manage data storage and retrieval from the database.
- **Hall Ticket and Barcode Generation Module:** This module will generate the halltickets in PDF format, embedding the Barcodes directly onto the tickets. It will alsoensure that each ticket meets the required formatting standards.

8. System Requirements:

❖ Software requirements:

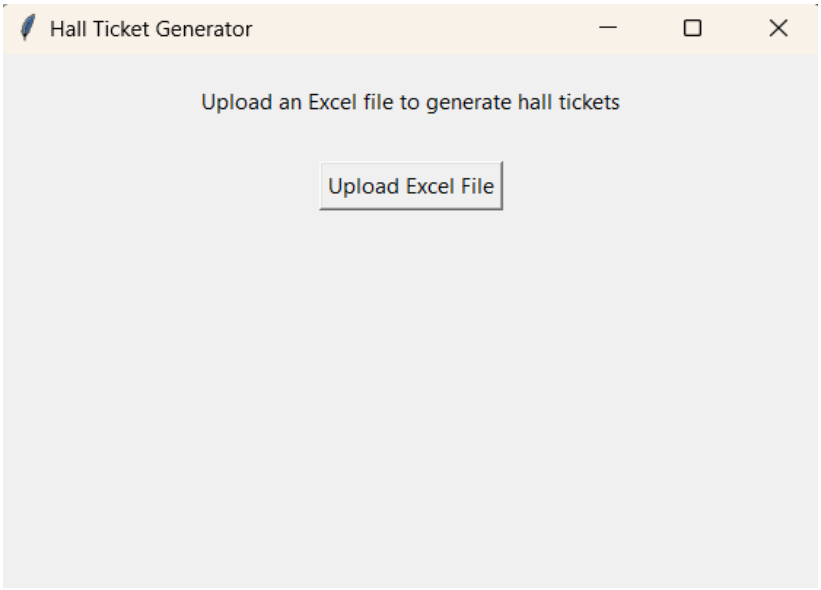
- Python 3.x: The primary programming language used for developing the system.
- Libraries: Essential Python libraries such as barcode for generating Barcodes, Pillow for image processing, ReportLab for PDF generation, and SQLite3or MySQL for database management.
- Operating System: The system can be developed and deployed on Windows, Linux, or macOS, depending on the institution's existing infrastructure.

❖ Hardware requirements:

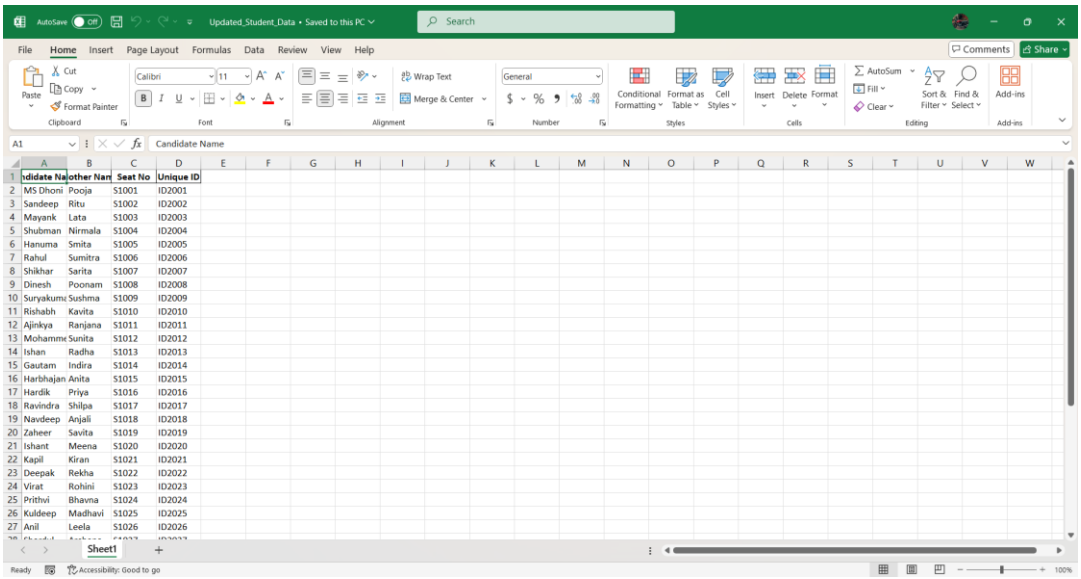
- A computer with at least 4 GB of RAM and a 2 GHz multi-core processor tohandle data processing and ticket generation efficiently.
- At least 500 MB of free storage for software installation and data storage.
- A scanner or mobile device with a camera for Barcode verification at theexamination centers.

9. Input/Output:

- Initially after running code we have to upload xlsx or csv file.
- A minimal graphical user interface (GUI) labeled "Hall Ticket Generator". Instruction displayed: "Upload an Excel file to generate hall tickets."
- Central button: Upload Excel File

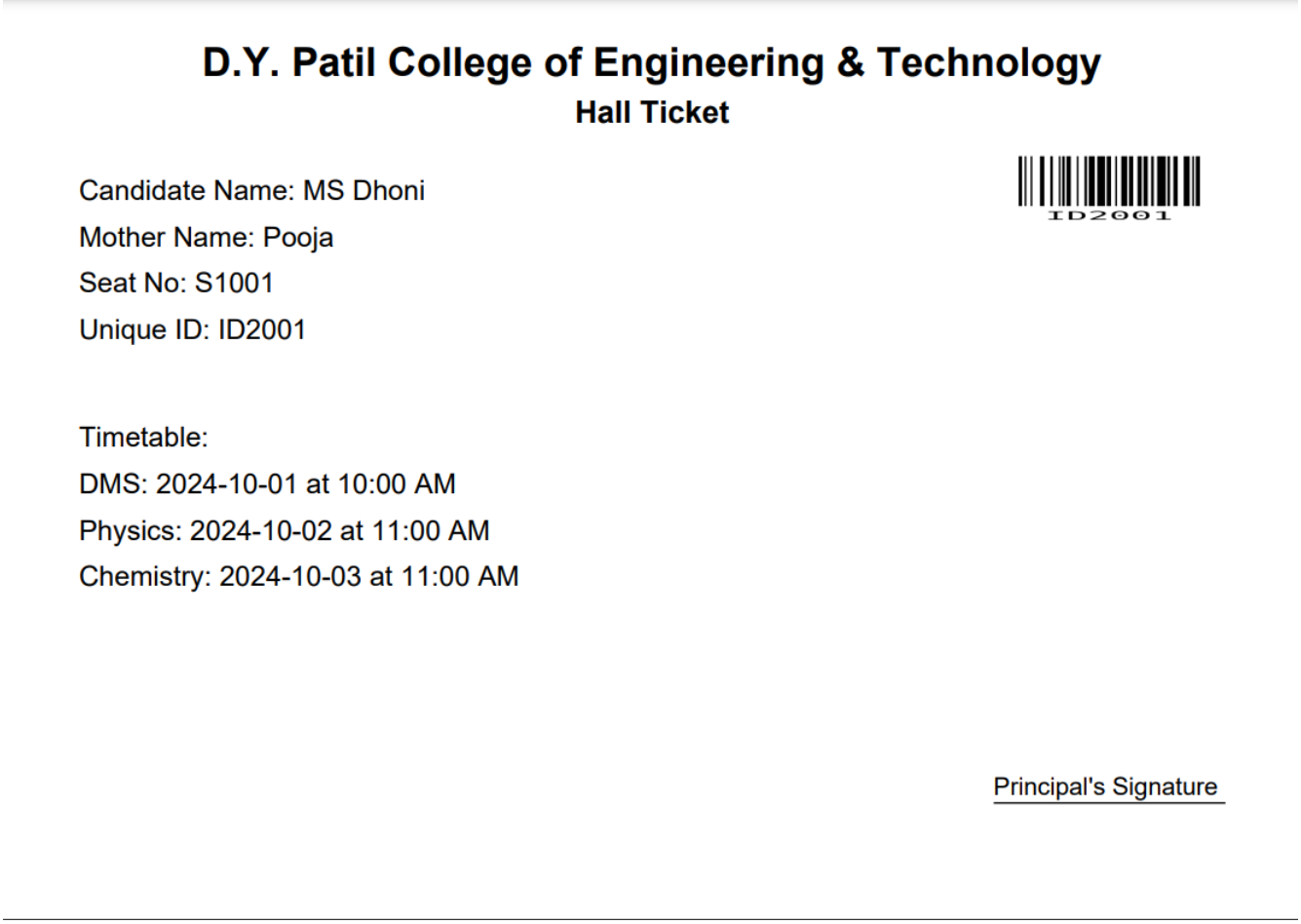


- Here excel file is already generated and added the required data.



didate Na	other Nan	Seat No	Unique ID
MS Dhoni	Pooja	S1001	ID2001
Sandeep	Ritu	S1002	ID2002
Mayank	Lata	S1003	ID2003
Shubman	Nirmala	S1004	ID2004
Hanuma	Smita	S1005	ID2005
Rahul	Sumitra	S1006	ID2006
Shikhar	Sarita	S1007	ID2007
Dinesh	Poonam	S1008	ID2008
Suryakum	Sushma	S1009	ID2009
Rishabh	Kavita	S1010	ID2010
Ajinkya	Ranjana	S1011	ID2011
Mohamme	Sunita	S1012	ID2012
Ishan	Radha	S1013	ID2013
Gautam	Indira	S1014	ID2014
Harbhajan	Anita	S1015	ID2015
Hardik	Priya	S1016	ID2016
Ravindra	Shilpa	S1017	ID2017
Navdeep	Anjali	S1018	ID2018
Zaheer	Savita	S1019	ID2019
Ishant	Meena	S1020	ID2020
Kapil	Kiran	S1021	ID2021
Deepak	Rekha	S1022	ID2022
Virat	Rohini	S1023	ID2023
Prithvi	Bhavna	S1024	ID2024
Kuldeep	Madhavi	S1025	ID2025
Anil	Leela	S1026	ID2026

- The screenshot showcases the output of a Hall Ticket Generator. It displays a clean and professional hall ticket design that includes the student's personal details, examination schedule, and a barcode for verification



D.Y. Patil College of Engineering & Technology

Hall Ticket

Candidate Name: MS Dhoni

Mother Name: Pooja

Seat No: S1001

Unique ID: ID2001

ID2001

Timetable:

DMS: 2024-10-01 at 10:00 AM

Physics: 2024-10-02 at 11:00 AM

Chemistry: 2024-10-03 at 11:00 AM

Principal's Signature

10. Conclusion:

The project "Hall Ticket Generator with Barcode" focuses on creating an efficient, secure, and scalable system to automate hall ticket generation for educational institutions. By integrating barcodes into the process, it enhances the security of student verification and reduces the risks of errors, duplication, and forgery. This project ultimately minimizes manual efforts and enhances the overall examination management process.

11. Reference

[1] Sarjiyus, O., Hamidu, M., & Audu, M. (2024). Simulation of Barcode Based Students' Examination Attendance System. European Journal of Theoretical and Applied Sciences, 2(4), 195-209.

[2] Yepuri, V. K., Pamu, G. C., Kodali, N., & Pradyumna, L. V. (2018). Examination Management Automation System. International Research Journal of Engineering and Technology (IRJET), 5(4), 2773-2779.

Class: TY **Div: A**

Group Member Name **Sign**

1. Soham Bagade
2. Swayam Dakare
3. Shridhar Badger
4. Ketan Patil

Date:

Place: Kolhapur

Sign of Guide	**Sign of Project Coordinator**	**Sign of HOD(CSE)**
(Prof. T. V. Salokhe)	(Prof. Nandini Patil)	(Prof. Ms. Raddhika. J. Dhannal)